

Drug repurposing for PCOS by drug-target interaction mining

Planning brief for the Big Data mini research proposal. Team reference, September 2026. This is the plan and the facts behind it, not the submitted text: each member writes their own proposal in their own words.

1. The assignment in one table

Section	What it must contain	Marks
Title	Disease, data, method in one line	3 (problem clarity and significance)
Biological problem	Unmet need with a number, the specific gap, the promised output. Half a page.	
Why big data	Which Vs apply, each with a size. The step that breaks a laptop.	2
Data	Real public sources, size, format, licence, one example record each	3 (data and approach)
Approach	Numbered pipeline, where Spark or distributed storage sits and why, a diagram	
Expected outcome	Concrete deliverable and a numeric success measure	2
Challenges	Hardest technical part and one honest limitation	

Length 2 to 3 pages. Hand-written or typed as instructed. Submitted through Google Classroom. The brief states that AI-generated text is detectable and receives no credit.

2. Why PCOS

Unmet need. Polycystic ovary syndrome affects roughly 1 in 10 women of reproductive age. No drug is approved specifically for PCOS. Treatment is symptomatic and borrowed: metformin (diabetes), combined oral contraceptives, spironolactone (hypertension), letrozole and clomiphene (fertility). Insulin resistance, hyperandrogenism and chronic low-grade inflammation are the three biological axes.

The gap. The borrowed drugs were found one at a time by clinical intuition. There is no systematic, evidence-weighted ranking of all approved drugs against the PCOS gene set. That ranking is the promised output.

Why repurposing. A new drug takes 10 to 15 years and costs over a billion dollars; an approved drug already has safety and pharmacokinetic data, so a repurposing lead can move to a trial in a fraction of the time.

The honest risk of PCOS as a choice. The disease gene set is small: about 19 to 20 GWAS loci (Day et al. 2018) and a few hundred associations in Open Targets, many with low evidence. The big-data argument therefore cannot rest on the disease side. It rests on the drug side (all bioactivity data), the network (the full human interactome) and, optionally, expression signatures.

3. Data sources

Source	Role	Approximate size	One record	Licence
ChEMBL v34/35	Drug-target bioactivities	2.4 M compounds, 20 M activities, 15 k targets. Dump ~4 GB compressed, ~25 GB loaded. Parquet on FTP.	CHEMBL25 (aspirin), target CHEMBL2094253 (PTGS1), IC50 = 32 nM, pchembl 7.5	CC BY-SA
DrugBank	Approved flag, indications	16.5 k drugs (2.8 k approved small molecules), 20 k drug-target links. XML 1.5 GB.	DB00331 metformin, approved, targets PRKAB1 (activator), indication: type 2 diabetes	Academic licence
BindingDB	Measured affinities, cross-check	2.9 M measurements, 1.3 M compounds, 9 k targets. TSV ~1 GB.	SMILES, UniProt P37231 (PPARG), Ki = 40 nM, PMID	CC BY 3.0
Open Targets Platform	PCOS-gene associations with evidence scores	63 k targets, 28 k diseases, 18 M evidence strings. Parquet, ~50 GB total; associations slice ~2 GB.	ENSG00000105329 (TGFB1), EFO_0000660 (PCOS), source ot_genetics_portal, score 0.42	CC0
GWAS Catalog	Genetic seed genes	700 k associations. TSV ~300 MB.	rs11031006, FSHB, polycystic ovary syndrome, p = 1e-12	Open
STRING v12, human	Protein interaction network	19.6 k proteins, 12 M scored links (1 M at score ≥ 700).	9606.ENSP00000284981, 9606.ENSP00000262607, combined_score 912	CC BY 4.0
LINCS L1000 (optional)	Drug-induced expression signatures	3 M profiles, 34 k perturbagens, 978 landmark genes. Level 5 ~100+ GB.	metformin, A549, 10 μ M, 978 z-scores	Open
GEO PCOS datasets (optional)	Disease expression signature	Granulosa cell and adipose microarray/RNA-seq sets, tens to hundreds of samples each.	GSE accession, sample, gene, log2 expression	Open
UniProt / Ensembl mapping	Identifier bridge	Human 20 k proteins; mapping files hundreds of MB.	P37231 ↔ ENSG00000132170 ↔ 9606.ENSP00000287820 ↔ PPARG	CC BY 4.0

ClinicalTrials.gov	Validation set	500 k trials.	NCT id, intervention metformin, condition PCOS, phase 3	Open
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Two corrections to the usual list. KEGG bulk download is licensed, so use Reactome for pathways. DisGeNET now restricts bulk access, so build the disease side on Open Targets and cite DisGeNET only as a cross-check. Stating this shows the veracity check was done.

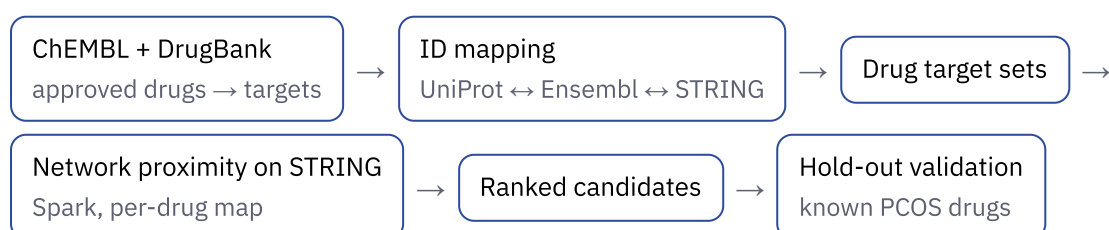
4. Why it is a big data problem, honestly

Step	Scale	Laptop?
Filter ChEMBL to approved drugs and join to targets	20 M activity rows in, ~55 k rows out	Input exceeds 8 GB RAM; needs a Spark join or a database. Output is small.
Build PCOS seed set from Open Targets + GWAS	2 GB Parquet in, ~300 to 500 genes out	Spark reads only the needed columns; result is small.
Network proximity for every approved drug on STRING	2,800 drugs × 12 M edges, degree-matched random controls (1,000 per drug)	No. ~2.8 million shortest-path computations. Embarrassingly parallel across drugs: Spark map over drug partitions with the graph broadcast.
Signature reversal against LINCS (optional)	3 M profiles, 100+ GB	No. Chunked or distributed processing required.

The four Vs, with numbers. Volume: 20 M activities, 12 M edges, optionally 3 M expression profiles. Variety: bioactivity tables, evidence strings, a graph, expression matrices, trial records, each with its own identifiers. Veracity: predicted versus experimental interactions, evidence scores from 0 to 1, licence-restricted sources excluded. Velocity: ChEMBL and Open Targets release quarterly, so the pipeline must re-run, which is an argument for automation, not for streaming.

Why Spark rather than Hadoop MapReduce. Network propagation and proximity are iterative; Spark keeps intermediate state in memory, MapReduce writes to disk each iteration. Open Targets already ships Parquet, Spark's native columnar format.

5. Pipeline



Open Targets + GWAS Catalog
PCOS seed genes with evidence scores

→ feeds proximity step

1. **Ingest.** Download ChEMBL (Parquet), DrugBank XML, Open Targets associations (Parquet), GWAS Catalog TSV, STRING human links. Store in HDFS or a local object store as Parquet. Tool: Spark.
2. **Approved drug set.** ChEMBL molecules with max_phase = 4, cross-checked against DrugBank "approved". Output ~2,800 drugs.
3. **Drug target sets.** ChEMBL activities with pchembl ≥ 6 (1 μ M) on human single-protein targets, joined to the approved set. Spark join over 20 M rows. Output: drug $\rightarrow$ set of UniProt accessions.
4. **PCOS seed set.** Open Targets associations for EFO_0000660 with overall score ≥ 0.1 , plus GWAS Catalog mapped genes. Weight = evidence score. Output ~300 to 500 Ensembl genes.
5. **Identifier mapping.** UniProt $\leftrightarrow$ Ensembl $\leftrightarrow$ STRING protein IDs. Log every unmapped ID; report the loss rate.
6. **Network proximity** (Guney et al. 2016). For each drug: mean shortest-path distance from its targets to the seed set on STRING (score ≥ 400), compared with 1,000 degree-matched random target sets, giving a z-score. Spark map over drugs, graph as broadcast variable.
7. **Rank and annotate.** Sort by z-score; attach for each drug its current indication, its closest PCOS genes and the evidence type behind them (the evidence path).
8. **Validate.** Before step 6, hold out drugs already used or in trials for PCOS (metformin, spironolactone, letrozole, clomiphene, pioglitazone, liraglutide, statins, inositol where listed). Measure recall of the hold-out in the top 50 and AUROC against all other approved drugs.
9. **Optional second line of evidence.** Signature reversal: PCOS granulosa or adipose signature from GEO versus LINCS drug signatures; report drugs that score on both lines.

6. Method options compared

Method	What it does	Strength	Weakness	Reference
Direct overlap (baseline)	Count of drug targets in the PCOS gene set, weighted by evidence	Trivially explainable	Only finds drugs on already-known genes	—
Network proximity (chosen)	Distance between targets and disease module versus random	Explainable, parallel, validated on many diseases	Favours well-studied proteins	Guney et al., Nat Commun 2016
Random walk with restart	Diffusion from seeds across the network	Captures multi-step effects	Iterative; parameter sensitive	Cowen et al., Nat Rev Genet 2017
Signature reversal	Drugs whose expression signature opposes the disease signature	Genuinely big data; independent evidence line	Cell lines are not ovary or adipose	Lamb 2006; Subramanian 2017
Knowledge-graph link prediction	Embedding model predicts drug-treats-disease edges	Powerful	Hard to explain in 3 pages; leakage risk	Himmelstein et al., eLife 2017

7. Expected outcome and success measure

Deliverable. A ranked table of approved drugs with proximity z-score, current indication, closest PCOS genes and evidence type, plus a network figure for the top 10.

Success measure. At least 4 of the roughly 8 held-out drugs already used for PCOS appear in the top 50 of about 2,800, against under 0.2 expected by chance; AUROC above 0.7. If the hold-out is not recovered, the negative result still shows whether PCOS biology is reachable through the current target space, which is itself informative.

8. Challenges and limitations

- **Identifier mapping** across five sources is the hardest engineering step; a 5 to 10% loss of targets is typical and must be reported.
- **Small, noisy disease gene set.** Mitigation: weight seeds by evidence type and test whether genetic seeds alone recover the hold-out better than all seeds.
- **Literature bias.** Well-studied proteins have more edges and more evidence, so proximity favours them. Mitigation: degree-matched randomisation, as in the chosen method.
- **Data leakage.** If known PCOS drugs stay in the seed-building step, the validation is circular. Mitigation: hold-out before scoring.
- **Honest limitation.** The output is a computational prioritisation; it needs experimental and clinical validation, and cell-line expression signatures, if used, do not represent ovarian tissue.

9. Possible expansions

Direction	What it adds
LINCS signature reversal	Second, independent evidence line; the largest data axis (100+ GB).
Evidence-type ablation	Tests whether genetic, expression or literature seeds drive the ranking; addresses veracity directly.
Side-effect filtering (SIDER, FAERS ~20 M reports)	Down-ranks candidates with adverse events relevant to women of reproductive age.
Tissue weighting (GTEx)	Weights targets by ovarian and adipose expression; cheap and biologically sensible.
Drug combinations	~4 M pairs of approved drugs; combinatorial, so genuinely distributed.
Docking of top hits	Structural plausibility check for the top 10 candidates against their PCOS-relevant targets.
Literature mining (PubTator, 36 M abstracts)	Published co-mentions supporting or contradicting each candidate.
Cross-disease run (PCOS and type 2 diabetes)	Shared insulin-resistance biology; turns the pipeline into a framework.

Knowledge-graph learning (Hetionet, PrimeKG)

The machine-learning extension, done properly.

Real-world evidence

Incidence in cohorts of patients already on a candidate drug; access-controlled, future work.

10. Writing checklist

- Every claimed V has a size next to it.
- The laptop-breaking step is named with a memory or compute reason.
- Each source has size, format, licence and one example record.
- The identifier-mapping step is written down.
- Known PCOS drugs are held out, not reported as discoveries.
- A success number and its chance baseline are stated.
- A diagram is drawn.
- Written in your own words; the reasoning must be yours to defend.

Dataset sizes are approximate, from 2025 releases. Cite the release you download.